A MORPHISM IS A GERBE EXACTLY WHEN BOTH IT AND ITS DIAGONAL ARE SURJECTIVE, FLAT AND OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The two clauses of the gerbe condition through local existence of objects and of isomorphisms. Used inside the counted proof (source lines 6532-6545). Supplies the local objects and isomorphisms from the flatness and finite presentation hypotheses.
- The statement that a morphism with surjective flat finitely presented diagonal and structure morphism has a flat finitely presented inertia. Invoked inside the counted proof (source lines 6546-6552). Connects the diagonal to the inertia.
- The criterion for flatness and local finite presentation along a flat finitely presented cover. Invoked inside the counted proof (source lines 6553-6560). Allows the gerbe condition to be checked after base change.

Source reference. Original lemma lemma-gerbe-descent: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6161–6161. Navigation pointer only: the counted text refers to the Stacks item lemma-gerbe-descent, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-gerbe-fppf: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6287–6287. Navigation pointer only: the counted text refers to the Stacks item lemma-gerbe-fppf, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-gerbe-isom-fppf: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6429–6429. Navigation pointer only: the counted text refers to the Stacks item lemma-gerbe-isom-fppf, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-gerbe-over-iso-classes: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6091–6091. Navigation pointer only: the counted text refers to the Stacks item lemma-gerbe-over-iso-classes, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-surjective-flat-locally-finite-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5877–5877. Navigation pointer only: the counted text refers to the Stacks item lemma-surjective-flat-locally-finite-presentation, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original proposition proposition-when-gerbe: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6319–6319. Navigation pointer only: the counted text refers to the Stacks item proposition-when-gerbe, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Proposition 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. The following are equivalent*

(1) *$\mathcal{X}$ is a gerbe over $\mathcal{Y}$, and*
(2) *$f : \mathcal{X} \to \mathcal{Y}$ and $\Delta : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ are surjective, flat, and locally of finite presentation.*

Proof. The implication (1) $\Rightarrow$ (2) follows from Lemmas lemma-gerbe-fppf and lemma-gerbe-isom-fppf.

Assume (2). It suffices to prove (1) for the base change of f by a surjective, flat, and locally finitely presented morphism $\mathcal{Y}' \to \mathcal{Y}$, see Lemma lemma-gerbe-descent (note that the base change of the diagonal of f is the diagonal of the base change). Thus we may assume $\mathcal{Y}$ is a scheme Y. In this case $\mathcal{I}_{\mathcal{X}} \to \mathcal{X}$ is a base change of Δ and we conclude that $\mathcal{X}$ is a gerbe by Proposition proposition-when-gerbe. We still have to show that $\mathcal{X}$ is a gerbe over Y. Let $\mathcal{X} \to X$ be the morphism of Lemma lemma-gerbe-over-iso-classes turning $\mathcal{X}$ into a gerbe over the algebraic space X classifying isomorphism classes of objects of $\mathcal{X}$. It is clear that $f : \mathcal{X} \to Y$ factors as $\mathcal{X} \to X \to Y$. Since f is surjective, flat, and locally of finite presentation, we conclude that $X \to Y$ is surjective as a map of fppf sheaves (for example use Lemma lemma-surjective-flat-locally-finite-presentation). On the other hand, $X \to Y$ is injective too: for any scheme T and any two T-valued points x_1, x_2 of X which map to the same point of Y, we can first fppf locally on T lift x_1, x_2 to objects ξ_1, ξ_2 of $\mathcal{X}$ over T and second deduce that ξ_1 and ξ_2 are fppf locally isomorphic by our assumption that $\Delta : \mathcal{X} \to \mathcal{X} \times_Y \mathcal{X}$ is surjective, flat, and locally of finite presentation. Whence $x_1 = x_2$ by construction of X. Thus $X = Y$ and the proof is complete. $\square$